

HOST (CPU)

Input

CAIDA / ER / BA topology -> A
MAWI trace -> $\lambda_i(t)$

Network-risk weighting

PageRank c_i , pilot var v_i ,
SLA s_i -> w_i (Eq. 10)

ANA allocation

$N_l \sim \sqrt{V_l w / C_l}$
weighted variance (Eq. 8-9)

*pilot per-node
variance v_i*

DEVICE (GPU)

Multi-GPU halo

METIS partition x G ranks
NCCL all-reduce every K steps

Coupled-SDE kernel

batched Philox randn, A C
Skorokhod reflect, 2-bucket EM step

MLMC hierarchy

levels $l = 0..L$, $h_l = h_0/2^l$
coupled fine/coarse, CRN

HOST (CPU)

Uncertainty-quantified output

mean queue occupancy + 95% CI
P95 / P99 tail delay
rare-event overflow P (IS)